Quantitative Macro Theory **HW #6**
Spring 2018 **Due at 5pm on *Friday* 3/23/2018**

Write your answer on latex and prepare a pdf file. The file should be clearly written and organized. You should think of that file as a short version of a paper. The appendix of that file should include your code. Email me that pdf file before the deadline.

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1. (50 points) **Households life-cycle problem**
   Calibration:

   - Individuals live between age $n = 25$ and $n = 80 = N$. The retire at $n = 65 = T$.
   - $u(c) = \frac{c^{1-\sigma}}{(1-\sigma)}$, with $\sigma = 2$.
   - The discount rate, $\beta = 0.92$.
   - Interest rate, $r = 0.02$.
   - Income is given by y, which has three components:

   $$y = \exp(z + f(n) + \varepsilon)$$

   - Use a three-state Markov chain Γ for the persistent component of income z, that we loosely identify with "blue-collar" ($z_1 = \log(15)$), "white-collar" ($z_2 = \log(80)$), and "rich" ($z_3 = \log(500)$).
   - Transition matrix:

   $$\begin{bmatrix} \Gamma_{1,1} = 0.60 & \Gamma_{1,2} = 0.35 & \Gamma_{1,3} = 0.05 \\ \Gamma_{2,1} = 0.10 & \Gamma_{2,2} = 0.80 & \Gamma_{2,3} = 0.10 \\ \Gamma_{3,1} = 0.05 & \Gamma_{3,2} = 0.35 & \Gamma_{3,3} = 0.60 \end{bmatrix}$$

   - The deterministic component of earnings is estimated using the PSID data. It has a hump-shaped profile where earnings grow by approximately 60% during the first 25 years and then decrease approximately by 18% until the end of the working life.

   $$f(n) = \log(-1.0135 + 0.1086 * n - 0.001122 * n^2).$$

   - And ε takes two values, {log(0.8),log(1.2)}, each with probability 1/2.
   - Finally, during retirement, which starts at n=66, their retirement income does not change and it is a function of the persistent component of income at age n=65:

   $$R(z) = 0.7 \times \exp(z_{65} + f(65)).$$

For $j \in \{low, high\}$, $i \in \{1, 2, 3\}$, and $n \in \{25, 26, ..., 64\}$ consider the problem

$$V(a, \varepsilon_j, z_i, n) = \max_{c,a'} u(c) + \beta \sum_{i'=1}^{3} \Gamma_{i,i'}[0.5V(a', \varepsilon_{low}, z_{i'}, n+1) + 0.5V(a', \varepsilon_{high}, z_{i'}, n+1)]$$

subject to

$$c + a'/(1+r) = a + \exp(z_i + f(n) + \varepsilon_j), \ a' \geq \underline{a},$$

for $n = 65$ note the same problem is slightly different

$$V(a, \varepsilon_j, z_i, 65) = \max_{c,a'} u(c) + \beta B(a', z_i, n+1)$$

subject to

$$c + a'/(1+r) = a + \exp(z_i + f(n) + \varepsilon_j), \ \boldsymbol{a' \geq 0}.$$

The retiree problem for $i \in \{1, 2, 3\}$ and $n \in \{66, 67, ..., 79\}$ is

$$B(a, z_i, n) = \max_{c,a'} u(c) + \beta B(a', z_i, n+1)$$

subject to

$$c + a'/(1+r) = a + R(z_i), \ \boldsymbol{a' \geq 0},$$

and the last period ($n = 80$) for $i \in \{1, 2, 3\}$ is simply

$$B(a, z_i, 80) = u((1+r)a + R(z_i)).$$

(a) (10 points) Set the lower bound of assets $\underline{a} = -100$. Create a grid for assets with equally spaced points for negative values and log scale for positive values. For the equally spaced part, you should have 101 points between $\underline{a}$ and 0. Add 50 points between $a = 1$ and $\bar{a} = 5000$ with log scale. You can do this by creating an auxiliary grid with equally spaced points for the variable b between $\log(1)$ and $\log(5000)$. Then obtain the grid for a doing for $i \in \{102, 103, ..., 151\}$ the following

$$a_i = \exp(b_i).$$

Solve for $B(a, z_i, 80)$ for all (a, z). How long did it take? Plot $B(., 80)$ as a function of a for the three values of z_i.

(b) (10 points) Solve for $B(a, z_i, 79)$ for all (a, z) using $B(a, z_i, 80)$ and doing the maximization using search in the discrete grid. Repeat the same exercises to solve for $B(a, z_i, 78)$ using $B(a, z_i, 79)$ and so on, until you solve $B(a, z_i, 66)$ using $B(a, z_i, 67)$. How long did it take? Plot the policy function for saving $A(., 66)$ as a function of a for the three values of z_i.

(c) (10 points) Solve for $V(a, \varepsilon_j, z_i, 65)$ for all (a, ε, z) using $B(a, z_i, 66)$ and doing the maximization using search in the discrete grid. Since the optimal a' is increasing in a, you can use some of the tricks we discussed in class to avoid checking every point.

Repeat the same exercises to solve for $V(a, \varepsilon_j, z_i, 64)$ using $V(a, \varepsilon_j, z_i, 65)$ and so on, until you solve $V(a, \varepsilon_j, z_i, 25)$ using $V(a, \varepsilon_j, z_i, 26)$. How long did it take? Plot the policy function for saving $A(., 25)$ as a function of a for z_1 and the two values of ε.

(d) (10 points) Simulate the paths for income, consumption, and savings for 1,000 individuals all starting at age $n = 25$ with $a = 0$, $z = z_1$, and $\varepsilon = \varepsilon_{low}$. Plot the mean of consumption, income, and savings with age in the x axis. Plot the variance of log(consumption) and log(income) and savings with age in the x axis.

(e) (10 points) Plot the share of agents with $a < 0$ (share of households in debt) with age in the x axis. Plot the mean of (-assets/income) conditional on the agents with $a < 0$ (this is the mean debt) with age in the x axis.

2. (50 points) **Role of Households Default**

Calibration: Use the same calibration that in question 1. There is only one new parameter, $\psi = 0.3$ (if you want, try changing this value and analyzing what happens with the plots in part (e) of this question).

We'll modify the model to add default. We will do it backward. *The retiree problem is the same that in question 1. We assume they cannot default. Note you can use the solution from that answer for that problem.*

For $n = 65$ note that the problem is slightly different because the household could default. If the household choose to pay, the value function is

$$V^P(a, \varepsilon_j, z_i, 65) = \max_{c,a'} u(c) + \beta B(a', z_i, n+1)$$

subject to

$$c + a'/(1+r) = a + \exp(z_i + f(n) + \varepsilon_j), \boldsymbol{a' \geq 0}.$$

Instead, if the household defaults, the value function is

$$V^D(\varepsilon_j, z_i, 65) = u(c) + \beta B(0, z_i, n+1)$$

subject to

$$c = (1-\psi)\exp(z_i + f(n) + \varepsilon_j).$$

Similarly for $n \in \{25, 26, ..., 64\}$ the value function for an agent that pays its debt is

$$V^P(a, \varepsilon_j, z_i, n) = \max_{c,a'} u(c) + \beta \sum_{i'=1}^{3} \Gamma_{i,i'}[0.5V(a', \varepsilon_{low}, z_{i'}, n+1) + 0.5V(a', \varepsilon_{high}, z_{i'}, n+1)]$$

subject to

$$c + a'q(a', z_i, n) = a + \exp(z_i + f(n) + \varepsilon_j).$$

and for an agent that default

$$V^D(\varepsilon_j, z_i, n) = u(c) + \beta \sum_{i'=1}^{3} \Gamma_{i,i'}[0.5V(0, \varepsilon_{low}, z_{i'}, n+1) + 0.5V(0, \varepsilon_{high}, z_{i'}, n+1)]$$

subject to

$$c = (1-\psi)\exp(z_i + f(n) + \varepsilon_j).$$

Note that there is a new policy function for $n \in \{25, 26, ..., 64\}$

$$\begin{aligned} P(a, \varepsilon_j, z_i, n) &= 1 \quad \text{if } V^P(a, \varepsilon_j, z_i, n) \geq V^D(\varepsilon_j, z_i, n) \\ &= 0 \qquad\qquad \text{otherwise.} \end{aligned}$$

And

$$V(a, \varepsilon_j, z_i, n) = \max\{V^P(a, \varepsilon_j, z_i, n), V^D(\varepsilon_j, z_i, n)\}.$$

We need to specify $q(a', z_i, n)$ for $n \in \{25, 26, ..., 65\}$. As in class, this is given by

$$q(a', z_i, n) = \frac{1}{1+r}\sum_{i'=1}^{3} \Gamma_{i,i'}[0.5P(a', \varepsilon_{low}, z_{i'}, n+1) + 0.5P(a', \varepsilon_{high}, z_{i'}, n+1)]$$

(a) (10 points) Use the same grid as in question 1. Solve for $V(a, \varepsilon_j, z_i, 65)$ for all (a, ε, z) using $B(a, z_i, 66)$ and doing the maximization using search in the discrete grid. Plot the policy function for default and repay $P(., 65)$ as a function of a for $a < 0$ given z_1 and the two values of ε in one figure. Plot the policy function for default and repay $P(., 65)$ as a function of a for $a < 0$ given z_2 and for the two values of ε in one figure.

(b) (10 points) Compute $q(a', z_i, 64)$ using the function $P(., 65)$. Plot the q function as a function of a for $a < 0$ given z_1. Do the same figure for z_2.

(c) (10 points) Solve for $V(a, \varepsilon_j, z_i, 64)$ using $q(a', z_i, 64)$ and $V(a, \varepsilon_j, z_i, 65)$. Do the same for $V(a, \varepsilon_j, z_i, 63)$ using $q(a', z_i, 63)$ and $V(a, \varepsilon_j, z_i, 64)$ and so on, until you solve $V(a, \varepsilon_j, z_i, 25)$ using $q(a', z_i, 25)$ and $V(a, \varepsilon_j, z_i, 26)$. How long did it take? Plot the policy function for repayment $P(., 25)$ for $a < 0$ as a function of a for z_1 and the two values of ε.

(d) (10 points) Simulate the paths for income, consumption, and savings for 1,000 individuals all starting at age $n = 25$ with $a = 0$, $z = z_1$, and $\varepsilon = \varepsilon_{low}$. Plot the mean of consumption for this economy together with the mean consumption in the economy in question 1 with age in the x axis. Do the same comparison for log(consumption).

(e) (10 points) Plot the share of agents with $a < 0$ (share of households in debt) with age in the x axis for this economy and the economy in question 1 in the same graph. Plot the mean of (-assets/income) conditional on the agents with $a < 0$ (this is the mean debt) with age in the x axis for this economy and the economy in question 1 in the same graph. For this economy, plot the share of agents that choose default for each age in a plot with age in the x axis.